An IC-PLS Framework for Group Corticomuscular Coupling Analysis

Xun Chen*, Chen He, Z. Jane Wang, *Senior Member, IEEE*, and Martin J. McKeown

***Abstract*—Corticomuscular coupling analysis, i.e., examining the relations between simultaneously recorded brain (e.g., electroencephalography—EEG) and muscle (e.g., electromyography—EMG) signals, is a useful tool for understanding aspects of human motor control. Traditionally, the most popular method to assess corticomuscular coupling has been the pairwise magnitude-squared coherence (MSC) between EEG and concomitant EMG recordings. In this paper, we propose assessing corticomuscular coupling by combining partial least squares (PLS) and independent component analysis (ICA), which addresses many of the limitations of MSC, such as difficulty in robustly assessing group inference and relying on the biologically implausible assumption of pairwise interaction between brain and muscle recordings. In the proposed framework, response relevance and statistical independence are jointly incorporated into a multiobjective optimization function to meaningfully combine the goals of PLS and ICA under the same mathematical umbrella. Simulations, performed under realistic assumptions, illustrated the utility of such an approach. The method was extended to address intersubject variability to robustly discover common corticomuscular coupling patterns across subjects. We then applied the proposed framework to concurrent EEG and EMG data collected in a Parkinson's disease (PD) study. The results from applying the proposed technique revealed temporal components in the EEG and EMG that were significantly correlated with one another. In addition to the expected motor areas, the corresponding spatial activation patterns demonstrated enhanced occipital connectivity in PD subjects, consistent with previous studies suggesting that PD subjects rely excessively on visual information to counteract the deficiency in being able to generate internal commands from their affected basal ganglia.**

***Index Terms*—Corticomuscular coupling, data fusion, electroencephalography (EEG), electromyography (EMG), group analysis, independent component analysis (ICA), Parkinson's disease (PD), partial least squares (PLS).**

I. Introduction

CORTICOMUSCULAR coupling analysis, i.e., studying simultaneous cortical and muscular activities typically during sustained isometric muscle contraction, is a key technique to assess functional interactions in the motor control system. The most common method to assess the interactions between motor-related brain areas and the muscles is the magnitude-squared coherence (MSC), which is a normalized measure of correlation between two waveforms or signals in the frequency domain. For example, in monkeys, coherent oscillations in the 20–30-Hz band can be detected between cortical local field potentials and the rectified electromyography (EMG) from contralateral hand muscles that are modulated during different phases of a precision grip task [1]. In humans, similar findings of beta-band corticomuscular coherences have also been detected during isometric contractions utilizing both magnetoencephalography [2] and electroencephalography (EEG) [3] recordings over the primary motor cortex.

Although MSC has been popular in studying corticomuscular coupling, it suffers from several limitations. First, incorporating intersubject variability in order to make a robust group inference is not straightforward with MSC because the exact frequency of maximum coupling may be inconsistent across subjects, necessitating *posthoc* pooled coherence analyses in an attempt to combine the original coherence estimates into a single representative estimate [4]. Second, the pairwise MSC concentrates on assessing the role of an individual locus in the brain in driving the motor system while motor activity is known to be more distributed [5]. In fact, recent work has suggested that *interactions between* brain regions correspond more closely to ongoing EMG than activity at discrete sites [6]. Moreover, when the brain activity is measured by EEG, applying MSC directly to raw EEG and EMG signals normally yields a very low coherence value, because only a small fraction of ongoing EEG activity is related to motor control [7]. Therefore, extensive statistical testing, with all the accompanying assumptions, is required to determine whether the EEG/EMG coherence is, in fact, significant.

However, several data-driven multivariate methods have been developed for analyzing biological data that may be appropriate for assessing corticomuscular coupling as they establish a dependency relationship between two types of data sets. These methods include multiple linear regression, principal component regression, partial least squares (PLS) and canonical correlation analysis (CCA) [8]. CCA and PLS rely on latent variables (LVs), which may enhance aid in the biological interpretation of the results. CCA is not as popular as PLS in practical applications [9], probably because often real-world biological data are high dimensional and collinear and, thus, applying CCA directly to the raw data may be ill conditioned. PLS exploits the covariation between predictor variables and response variables and extracts a new set of latent components that maximally relate them [9].

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*X. Chen is with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: xunchen@ece.ubc.ca).

C. He and Z. J. Wang are with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: chenh@ece.ubc.ca; zjanew@ece.ubc.ca).

M. J. McKeown is with the Pacific Parkinsons Research Centre, Department of Medicine (Neurology), University of British Columbia, Vancouver, BC V6T 2B5, Canada (e-mail: martin.mckeown@ubc.ca).

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PLS and its variants have been investigated in many medical applications, such as assessing the spatial patterns of brain activity in functional magnetic resonance imaging (fMRI) data associated with behavioral measures [10], and the common temporal components between EEG and fMRI signals [11]. In addition to the ability of handling high dimensional and collinear data, PLS is sufficiently flexible that it can be extended to perform group level analysis and to accommodate multiway data, such as 3-D time-frequency-EEG signals incorporating joint temporal, spectral, and spatial decompositions [6]. However, PLS can only extract uncorrelated LVs and their interpretations may be difficult in real applications [12].

Independent component analysis (ICA) is based on the notion that it is insufficient to consider only up to the second-order statistics (e.g., correlation and covariance) for obtaining a unique LV model [13] if the data are not strictly multivariate gaussian. ICA assumes that the multivariate data are composed of a linear superposition of mutually statistically independent signal sources. In statistics, independence is a much stronger condition than uncorrelatedness, and thus ICA algorithms typically employ criteria related to information theory and/or negentropy. ICA has found many potential biomedical applications [14]–[16]. For example, ICA can be used to reliably separate fMRI data into meaningful constituent components, including consistently and transiently task-related physiological changes, non-task-related physiological phenomena, and machine or movement artifacts [14].

Independent component regression (ICR) has been developed as an alternative to traditional LV-based methods (e.g., PLS) [17]. In ICR, ICs are first extracted from the measurements and then linear regression is performed to relate the ICs with the response. However, as a two-stage method, ICR inherits the following drawbacks of ICA: First, ICA can only decompose one data set at a time and, thus, ignores the effects of the response variables. Even though the extracted ICs are maximally independent with each other, they may, and are typically not, directly informative about the response. Second, unlike principal component analysis, which yields unique and ranked principal components (PCs) by best explaining the variance in the data, classical ICA with stochastic learning rules can result in ICs that vary by permutation and/or dilation even when repeatedly performed on the same data set. Such inherent indeterminacy of classical ICA is due to the fact that the assumptions are mathematically insufficient to extract the independent sources exactly in their original form. To improve the performance of ICR, a modified ICR has been proposed, in which the order of the components extracted from the measured data $\boldsymbol{X}$ could be properly determined according to the response $\boldsymbol{Y}$ [18]. However, this approach only decomposes $\boldsymbol{X}$ but not $\boldsymbol{Y}$.

Thus, both PLS and ICA have their advantages and disadvantages, but perhaps most importantly, these two methods can be considered complementary. In this paper, we propose combining their advantages and minimizing their drawbacks by formulating a multiobjective optimization problem to simultaneously incorporate response relevance and independence into the regression procedure, to produce a so-called IC-PLS model. The proposed IC-PLS extracts LVs from both the measured data $\boldsymbol{X}$ and the response $\boldsymbol{Y}$, keeping the LVs maximally independent and uniquely sorting the LVs in order of relevance. When applied to corticomuscular coupling analysis, the proposed IC-PLS can extract the most significant LV pairs from concurrent EEG and EMG data in an orderly manner. Furthermore, to infer consistent activation patterns across subjects within a group in the face of intersubject variability, we also develop a group analysis framework based on the proposed IC-PLS model to accommodate the multisubject case and achieve robust group inference.

We evaluated the performance of the proposed method. We first validated the proposed IC-PLS model on simulated data. We then applied the proposed group analysis framework to concurrent EEG and EMG data collected from patients with Parkinson's disease (PD) and age-matched normal subjects as they performed a dynamic, visually guided tracking task. We found highly correlated temporal patterns between EEG and EMG signals and meaningful spatial activation patterns. While the proposed method is intentionally proposed for corticomuscular coupling analysis, we note it can also be applied to analyze other types of concurrent signals, including, but not limited to, fMRI, photoplethysmograph (PPG), electrocardiography (ECG), and kinematic data.

II. Methods

A. IC-PLS Model

In this section, we first formulate the corticomuscular coupling analysis problem as a multiobjective optimization problem. The basic idea is to extract highly correlated, but still maximally independent components in both the EEG and EMG so that the covariance between each of the extracted EEG and EMG components is maximized. Since the components are a result of an optimization based on a weighted combination of statistical independence and response-similarity goals, we then design some strategies to adjust each subobjective's parameters simultaneously and make the corresponding subobjectives change in parallel. Because of the importance of the initial solution, we also describe the initialization setting procedure.

1) Multiobjective Optimization: To combine the advantages of PLS and ICA and design an overall optimization objective, the following conditions should be satisfied simultaneously:

First, PLS exploits the covariation between predictor variables and response variables and tries to find a new set of LVs that maximally relate them [9]. In other words, the covariance between the extracted LVs should be maximized as

$$\max_{\boldsymbol{w}_1,\boldsymbol{w}_2} \quad \left(\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\right)^2 \qquad \text{s.t.} \quad \boldsymbol{w}_i{}^T\boldsymbol{w}_i = 1, \quad i = 1, 2 \tag{1}$$

where $\boldsymbol{w}_i$s $(i = 1, 2)$ are the weight vectors, $\boldsymbol{x}$ is the predictor vector with size $p \times 1$, and $\boldsymbol{y}$ is the response vector with size $q \times 1$. Here, it is assumed that $\boldsymbol{x}$ and $\boldsymbol{y}$ are preprocessed (the specific procedure will be presented in Section II-A3).

Second, to incorporate the advantages of ICA, the extracted LVs should be as independent with each other as possible. According to Hyvärinen and Oja [13], this can be achieved by

solving the following problems:

$$\max_{\boldsymbol{w}_1} \quad \left(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1))\right)^2 \quad \text{s.t.} \quad \boldsymbol{w}_1{}^T\boldsymbol{w}_1 = 1 \tag{2}$$

and

$$\max_{\boldsymbol{w}_2} \quad \left(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y})) - \mathrm{E}(\mathrm{G}(u_2))\right)^2 \quad \text{s.t.} \quad \boldsymbol{w}_2{}^T\boldsymbol{w}_2 = 1 \tag{3}$$

where u_1 and u_2 are standard Gaussian variables and $\mathrm{G}(\cdot)$ is a nonquadratic function. The following choices of $\mathrm{G}(\cdot)$ have been previously suggested:

$$\mathrm{G}_1(u) = \frac{1}{a_1}\log\cosh(a_1 u) \quad \text{and} \quad \mathrm{G}_2(u) = -\exp(-\frac{u^2}{2}) \tag{4}$$

where the constant a_1 is generally $1 \le a_1 \le 2$. In this paper, we adopt $\mathrm{G}_1(\cdot)$ for $\mathrm{G}(\cdot)$ in the following optimization procedure.

To encapsulate the above three maximization objectives, an intuitive approach is to combine the three objectives into a single aggregate objective function and achieve a good tradeoff between their respective goals. We, therefore, employ the well-known weighted linear sum of the subobjectives, and propose the following multiobjective optimization problem:

$$\begin{aligned}\max_{\boldsymbol{w}_1,\boldsymbol{w}_2} \quad & \alpha\left[\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\right]^2 \\ & + \beta\left[\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1))\right]^2 \\ & + \theta\left[\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y})) - \mathrm{E}(\mathrm{G}(u_2))\right]^2 \\ \text{s.t.} \quad & \boldsymbol{w}_i{}^T\boldsymbol{w}_i = 1, \quad i = 1,2\end{aligned} \tag{5}$$

where α, β, and θ are the weights for the corresponding subobjectives, respectively, and satisfy $\alpha + \beta + \theta = 1$. In Section II-A2, we will further discuss how to adjust these parameters in detail.

Based on the above optimization formulation, a solution can be calculated using an approximate Newton iteration approach (with the detailed derivation being shown in Appendix). At each iteration, $\boldsymbol{w}_i$s are updated as

$$\begin{aligned}&\boldsymbol{w}_i \leftarrow \boldsymbol{w}_i + \boldsymbol{d}_i, \\ &\text{i.e.,} \quad \boldsymbol{w}_i \leftarrow \boldsymbol{w}_i - (\mathrm{J}\Phi(\boldsymbol{w}_i))^{-1}\nabla\mathrm{F}_{\boldsymbol{w}_i}, \quad i = 1,2\end{aligned} \tag{6}$$

where $\mathrm{J}\Phi(\boldsymbol{w}_i)$ is the Jacobian matrix and $\nabla\mathrm{F}_{\boldsymbol{w}_i}$ is the first-order derivative of $\mathrm{F}(\boldsymbol{w}_1, \boldsymbol{w}_2, \lambda_1, \lambda_2)$ defined in (A-1) with respect to $\boldsymbol{w}_i$.

2) Determining the Optimization Weights: The primary goal during the multiobjective optimization is to simultaneously optimize two or more separate objectives subject to certain constraints. Due to the complicated and possibly conflicting nature of the subobjectives, the overall optimal solution obtained depends on the relative values of the weights specified (i.e., α, β, and θ here). Therefore, to achieve a good tradeoff between different subobjectives, it is important to determine the weights appropriately. Here, two key issues should be taken into consideration: the different scales and the different convergence speeds of the subobjectives [19].

The following strategy was employed to avoid the undesirable situation in which one subobjective overwhelms the others: the aforementioned weights α, β, and θ are decomposed into three factors, respectively: $\alpha_{\mathrm{sig}} \cdot \alpha_{\mathrm{scale}} \cdot \alpha_{\mathrm{adj}}, \beta_{\mathrm{sig}} \cdot \beta_{\mathrm{scale}} \cdot \beta_{\mathrm{adj}}$, and $\theta_{\mathrm{sig}} \cdot \theta_{\mathrm{scale}} \cdot \theta_{\mathrm{adj}}$. Here, $\alpha_{\mathrm{sig}}, \beta_{\mathrm{sig}}$, and θ_{sig}, called significance factors, denote the relative significance attached to each subobjective. They can be set subjectively according to the specific application. $\alpha_{\mathrm{scale}}, \beta_{\mathrm{scale}}$, and θ_{scale}, named scale factors, are adopted to unify the scales of the three subobjectives. Just as their names imply, the scale factors are defined as follows:

$$\alpha_{\mathrm{scale}} = \frac{1}{|\mathrm{F}_{1,\boldsymbol{w}_1\boldsymbol{w}_2}|}, \ \beta_{\mathrm{scale}} = \frac{1}{|\mathrm{F}_{2,\boldsymbol{w}_1}|}, \ \theta_{\mathrm{scale}} = \frac{1}{|\mathrm{F}_{3,\boldsymbol{w}_2}|} \tag{7}$$

where the denominators are the absolute values of the corresponding subobjective functions defined as

$$\begin{aligned}\mathrm{F}_{1,\boldsymbol{w}_1\boldsymbol{w}_2} &= \left(\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\right)^2 + \kappa_{11}(\boldsymbol{w}_1{}^T\boldsymbol{w}_1 - 1) \\ &\quad + \kappa_{12}(\boldsymbol{w}_2{}^T\boldsymbol{w}_2 - 1) \\ \mathrm{F}_{2,\boldsymbol{w}_1} &= \left(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1))\right)^2 + \kappa_2(\boldsymbol{w}_1{}^T\boldsymbol{w}_1 - 1) \\ \mathrm{F}_{3,\boldsymbol{w}_2} &= \left(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y})) - \mathrm{E}(\mathrm{G}(u_2))\right)^2 + \kappa_3(\boldsymbol{w}_2{}^T\boldsymbol{w}_2 - 1)\end{aligned} \tag{8}$$

where $\kappa_{11}, \kappa_{12}, \kappa_2$, and κ_3 are Lagrange multipliers. $\alpha_{\mathrm{adj}}, \beta_{\mathrm{adj}}$, and θ_{adj}, called adjustable factors, are used to balance their different convergence speeds and can be updated in real time during the iterative searching procedure. In general, the gradient of a function can be employed to estimate the speed of convergence. Therefore, the adjustable factors can be defined as below

$$\begin{aligned}\alpha_{\mathrm{adj}} &= \frac{1}{(\|\nabla\mathrm{F}_{1,\boldsymbol{w}_1}\| + \|\nabla\mathrm{F}_{1,\boldsymbol{w}_2}\|)/2} \\ \beta_{\mathrm{adj}} &= \frac{1}{\|\nabla\mathrm{F}_{2,\boldsymbol{w}_1}\|}, \ \theta_{\mathrm{adj}} = \frac{1}{\|\nabla\mathrm{F}_{3,\boldsymbol{w}_2}\|}\end{aligned} \tag{9}$$

where the denominators are the L^2 norm of the corresponding first-order derivatives, which are calculated in a similar way to the calculation of $\nabla\mathrm{F}_{\boldsymbol{w}_i}\,(i = 1,2)$ in Appendix

$$\begin{aligned}\nabla\mathrm{F}_{1,\boldsymbol{w}_1} &= 2\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y})) \\ &\quad - 2\boldsymbol{w}_1{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\boldsymbol{w}_1 \\ \nabla\mathrm{F}_{1,\boldsymbol{w}_2} &= 2\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{y}(\boldsymbol{w}_1{}^T\boldsymbol{x})) \\ &\quad - 2\boldsymbol{w}_2{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{y}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\boldsymbol{w}_2 \\ \nabla\mathrm{F}_{2,\boldsymbol{w}_1} &= 2(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x})) \\ &\quad - 2\boldsymbol{w}_1{}^T(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\boldsymbol{w}_1 \\ \nabla\mathrm{F}_{3,\boldsymbol{w}_2} &= 2(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y})) - \mathrm{E}(\mathrm{G}(u_2)))\mathrm{E}(\boldsymbol{y}\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y})) \\ &\quad - 2\boldsymbol{w}_2{}^T(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y})) - \mathrm{E}(\mathrm{G}(u_2)))\mathrm{E}(\boldsymbol{y}\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\boldsymbol{w}_2\end{aligned} \tag{10}$$

where $\mathrm{g}(\cdot)$ represents the first-order derivative of $\mathrm{G}(\cdot)$. From the aforesaid definitions, we can see that $\alpha_{\mathrm{adj}}, \beta_{\mathrm{adj}}$, and θ_{adj} will be adjusted online during the optimization procedure. If one subobjective changes faster, i.e., its gradient norm is larger, the

corresponding adjustable factor will be smaller and vice versa. This makes all subobjectives change in parallel. However, when the iteration results are close to the optima, the gradients will approach zero. Therefore, we set the three weights as below

$$\begin{cases} \alpha = \alpha_{\text{sig}} \cdot \alpha_{\text{scale}} \cdot \alpha_{\text{adj}}, & \alpha_{\text{adj}} \leq \text{TH} \\ \beta = \beta_{\text{sig}} \cdot \beta_{\text{scale}} \cdot \beta_{\text{adj}}, & \beta_{\text{adj}} \leq \text{TH} \\ \theta = \theta_{\text{sig}} \cdot \theta_{\text{scale}} \cdot \theta_{\text{adj}}, & \theta_{\text{adj}} \leq \text{TH} \\ \alpha = \alpha_{\text{sig}} \cdot \alpha_{\text{scale}}, & \alpha_{\text{adj}} > \text{TH} \\ \beta = \beta_{\text{sig}} \cdot \beta_{\text{scale}}, & \beta_{\text{adj}} > \text{TH} \\ \theta = \theta_{\text{sig}} \cdot \theta_{\text{scale}}, & \theta_{\text{adj}} > \text{TH} \end{cases} \tag{11}$$

where TH is a predefined small threshold that envelops a narrow range around the optimum. Whenever the searching route enters the range, the corresponding gradient approximates zero so that the weights should be determined only by the first two factors. In short, the final weights take into account the effects of all the three factors.

3) Initialization Setting: When solving optimization problems, the final solutions not only depend upon the search algorithm employed, but also upon the initial setting, which determines the searching path and starting point, respectively. In the proposed IC-PLS model, to prevent the optimization being stuck in an uninformative local minimum, we introduce a joint statistical analysis method [20], [21] and use the extracted LVs as initial conditions for the subsequent iterative search.

Suppose the original predictor and response data are denoted by $\boldsymbol{Z}_1$ (with size $N \times P$) and $\boldsymbol{Z}_2$ (with size $N \times Q$), respectively, and define the sublatent variables (subLVs) in each data space to be linear combinations of the original variables, i.e., $\boldsymbol{Z}_1\boldsymbol{v}_1, \boldsymbol{Z}_2\boldsymbol{v}_2$. One super latent variable (supLV) $\boldsymbol{t}_g$ is designed to relate the subLVs by solving the following optimization problem:

$$\begin{aligned} &\max\, (\boldsymbol{t}_g{}^T \boldsymbol{Z}_1\boldsymbol{v}_1)^2 + (\boldsymbol{t}_g{}^T \boldsymbol{Z}_2\boldsymbol{v}_2)^2 \\ &\text{s.t.}\, \boldsymbol{t}_g{}^T\boldsymbol{t}_g = 1, \quad \boldsymbol{v}_i{}^T\boldsymbol{v}_i = 1, \quad \forall i = 1, 2 \end{aligned} \tag{12}$$

with all data assumed to be zero mean and normalized to unit variance in advance. The subLV $\boldsymbol{Z}_i\boldsymbol{v}_i$ in each data space carries its associated variation information and $(\boldsymbol{t}_g{}^T \boldsymbol{Z}_i\boldsymbol{v}_i)^2$ models the covariance information between the subLV $\boldsymbol{Z}_i\boldsymbol{v}_i$ and the supLV $\boldsymbol{t}_g$. The supLV $\boldsymbol{t}_g$ relates all subLVs simultaneously and actually plays a role as the link bridge. It is expected that, by solving the above optimization problem, the extracted subLVs will carry as much variation as possible in each data space and at the same time be correlated as closely as possible.

By the method of Lagrange multipliers, the supLV and subLVs can be readily derived as

$$(\boldsymbol{Z}_1\boldsymbol{Z}_1{}^T + \boldsymbol{Z}_2\boldsymbol{Z}_2{}^T)\boldsymbol{t}_g = \rho_g\boldsymbol{t}_g \tag{13}$$

$$\boldsymbol{t}_i = \boldsymbol{Z}_i\boldsymbol{v}_i = \sqrt{\frac{1}{\rho_i}}\boldsymbol{Z}_i\boldsymbol{Z}_i{}^T\boldsymbol{t}_g \tag{14}$$

where $\rho_i = \boldsymbol{t}_g{}^T\boldsymbol{Z}_i\boldsymbol{Z}_i{}^T\boldsymbol{t}_g$. Thus, with Lagrange multipliers the optimization problem is now characterized as a standard algebra problem.

Several supLVs, represented by $\boldsymbol{T}_g$ and ranked by descending ρ_gs can be extracted and a same number of subLVs, represented by $\boldsymbol{T}_i$, can be readily calculated by using (14)

$$\boldsymbol{T}_i = \boldsymbol{Z}_i\boldsymbol{Z}_i{}^T\boldsymbol{T}_g\boldsymbol{D}_i \tag{15}$$

where $\boldsymbol{D}_i$ is a diagonal matrix with corresponding $\sqrt{\frac{1}{\rho_i}}$s as its diagonal elements. In practice, we need to specify the number of supLVs to be extracted, which can be done by extracting a sufficient number to explain an adequate fraction of the variance (e.g., 90%).

From the aforesaid derivation, it is apparent that the supLVs $\boldsymbol{T}_g$ are actually the PCs of the concatenated data shown in (13) and the subLVs $\boldsymbol{T}_i$ are extracted by regressing each data space on the supLVs as shown in (15). Hence, the supLVs can represent the general systematic variations for both data spaces and the subLVs can capture the local systematic variations within each data space. However, collinearity may exist in the subLVs calculated through the previous procedure because each data set is used repetitively as shown in (14). Furthermore, the subLVs are not sorted according to the correlation of the corresponding between-set subLV pairs. To efficiently implement the subsequent Newton iteration, we need to address the above concerns and, therefore, modify the original subLVs as follows:

First, find the maximal correlation and the corresponding supLV $\boldsymbol{t}_g^*$ as well as subLVs $\boldsymbol{t}_1^*, \boldsymbol{t}_2^*$;
Second, exclude $\boldsymbol{t}_g^*$ from $\boldsymbol{T}_g$ and update each data space using $\boldsymbol{t}_1^*, \boldsymbol{t}_2^*$ in a deflation manner;
Third, update the original subLVs based on the new supLVs and the updated data spaces by using (15).

After repeating these steps, it is ensured that the modified subLVs (MsubLVs) are orthogonal to each other and automatically ordered. The extracted MsubLVs are then used as the initial solutions for the proposed multiobjective optimization problem. (A MATLAB package for the IC-PLS model is available by contacting Xun Chen at xunchen@ece.ubc.ca).

B. Group Analysis Framework for Corticomuscular Activity Analysis

In many medical applications, the desire to make group inferences requires recruitment of an adequate number of subjects and the application of a suitable group analysis, which accommodates intersubject variability. Here, we propose a group analysis framework for modeling corticomuscular coupling activity, and apply the proposed group analysis framework to the EEG/EMG data collected from normal subjects and subjects with PD during a motor task.

1) Group Analysis Framework: As illustrated in Fig. 1, there are two stages in this framework. Suppose we have B subjects in total (e.g., $B = 17$ in this study) and two types of data sets for each subject. In the first stage, the subLVs $\boldsymbol{T}_{bj}$'s ($b = 1, 2, \ldots, B$; $j = 1, 2$) for each subject are calculated by using (13) and (15). Note that these subLVs include most of the useful information for further analysis, and there is no need to use the modified algorithm here because the collinearity and ordering problems will be solved together in the initialization step of the group level IC-PLS model.

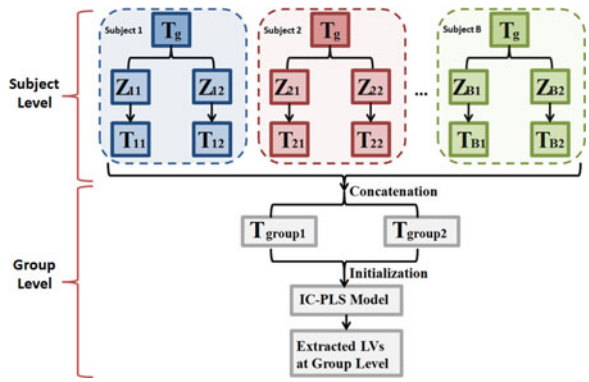

Fig. 1. Diagram of the group analysis framework based on the proposed IC-PLS model.

In the second stage, all subjects' subLVs are correspondingly concatenated as

$$\boldsymbol{T}_{\text{groupj}} = [\boldsymbol{T}_{1j}, \boldsymbol{T}_{2j}, \ldots, \boldsymbol{T}_{Bj}], \quad \forall j = 1, 2 \tag{16}$$

with the assumption that all subjects share common temporal group patterns [16]. Then, the whole initialization procedure described in Section II-A3 is applied to the concatenated data $\boldsymbol{T}_{\text{groupj}}$. Finally, the proposed IC-PLS model is employed to process the initialized data sets and the expected LVs could be extracted to represent group patterns.

2) Feature Extraction: In most existing studies, the corticomuscular coupling analysis is performed directly on the raw EEG and EMG data, which typically yields small coherence values. Nonetheless, with proper preprocessing steps, highly correlated EEG and EMG feature(s) can be extracted from the raw signals. In this study, we examined the coupling relationships between time-varying EEG features and amplitudes of the EMG signals, constituting $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$, respectively, for each subject.

EEG Features: Two types of EEG features were considered: pairwise Pearson's correlation [22] and band-limited, pairwise EEG coherences [23]. Pearson's correlation measures the dependencies between a pair of EEG signals $\boldsymbol{e}^* = (e_1^*, e_2^*, \ldots, e_n^*)$ and $\boldsymbol{e}^\circ = (e_1^\circ, e_2^\circ, \ldots, e_n^\circ)$ in the time domain as below

$$\gamma_{e^*e^\circ} = \frac{\sum_{i=1}^{n}(e_i^* - \bar{e^*})(e_i^\circ - \bar{e^\circ})}{\sqrt{\sum_{i=1}^{n}(e_i^* - \bar{e^*})^2 \sum_{i=1}^{n}(e_i^\circ - \bar{e^\circ})^2}} \tag{17}$$

where $\bar{e^*}$ and $\bar{e^\circ}$ are the sample means of $\boldsymbol{e}^*$ and $\boldsymbol{e}^\circ$. EEG coherence can also be used to examine the relation between two EEG signals, which includes both power and frequency information

$$Coh(f) = |\boldsymbol{S}_{e^*e^\circ}(f)|^2 / (\boldsymbol{S}_{e^*e^*}(f)\boldsymbol{S}_{e^\circ e^\circ}(f)) \tag{18}$$

where $|\boldsymbol{S}_{e^*e^\circ}|$ is the cross spectrum between the signals $\boldsymbol{e}^*$ and $\boldsymbol{e}^\circ$; $\boldsymbol{S}_{e^*e^*}$ is the autospectrum of the signal $\boldsymbol{e}^*$; $\boldsymbol{S}_{e^\circ e^\circ}$ is the autospectrum of the signal $\boldsymbol{e}^\circ$. In this study, we calculated both the time-varying pairwise correlations and coherences between EEG channels, using a Hamming window with length 300 and with a 95% overlap. To be consistent with common EEG practice, we further divide the coherence into four frequency bands, i.e., theta band (4–8 Hz), alpha band (8–13 Hz), beta band (13–30 Hz), and gamma band (30–70 Hz). The time-varying energy signals in each band were then used as the EEG features. Therefore, the raw EEG information can be represented by a 2-D matrix with size $N \times M$, where the rows correspond to the samples at different time points and the columns correspond to the features, i.e., pairwise correlations or band-limited coherences between the EEG channels.

EMG Features: An individual EMG channel signal can be considered as a zero-mean, band-limited, and wide-sense stationary stochastic process modulated by the EMG amplitude, which represents the overall muscle activity of individual underlying muscle fibers [24]. While different techniques have been proposed for accurate amplitude estimation, in this study, we employ the root-mean-square (RMS) approach to calculate the EMG amplitude of short-duration EMG signals $\boldsymbol{e} = (e_1, e_2, \ldots, e_n)$

$$e_{\text{rms}} = \sqrt{\frac{1}{n}({e_1}^2 + {e_2}^2 + \cdots + {e_n}^2)}. \tag{19}$$

A moving window with length $n = 300$ and a 95% overlap was applied here, the same as in the EEG correlation calculation, to ensure that the obtained EEG and EMG features were temporally aligned and matched.

3) Significance Assessment: To determine the statistical significance levels of the extracted LVs, we utilized a nonparametric permutation test [25] in which the temporal order of EEG features $\boldsymbol{Z}_1$ was uniformly permuted for all subjects while keeping the EMG features $\boldsymbol{Z}_2$ unchanged. Two hundred random permutations were generated. The proposed group analysis framework described in Section II-B1 was applied to each of these permutations. The correlation coefficients among the extracted temporal patterns from permuted EEG features and unchanged EMG features were then calculated to form an empirical null distribution. The p-value of the original EEG/EMG correlation was then computed from the null distribution as the proportion of sampled permutations whose correlations were greater than or equal to the original correlation. The components with p-value being less than 0.05 were considered to be statistically significant, denoted as LV_{EEG} and LV_{EMG}, both with size $(N \times K)$, where K is the number of significant components.

4) Spatial Pattern Extraction: Our goal was to investigate the differences in spatial patterns of EEG channels between the normal and PD patient groups while subjects performed a motor task. After the identification of significant temporal patterns, we can regress the EEG-related components LV_{EEG} back onto the EEG features $\boldsymbol{Z}_{b1}$ (for $b = 1, 2, \ldots, B$) for all subjects as below

$$\boldsymbol{p}_{bk} = \sqrt{\frac{1}{\boldsymbol{lv}_k^T \boldsymbol{Z}_{b1} {\boldsymbol{Z}_{b1}}^T \boldsymbol{lv}_k}} {\boldsymbol{Z}_{b1}}^T \boldsymbol{lv}_k, \quad k = 1, 2, \ldots, K \tag{20}$$

where $\boldsymbol{lv}_k$ is the kth column of LV_{EEG} and $\boldsymbol{p}_{bk}$ is the spatial pattern of the kth component for subject b. In addition, we also want to determine which EEG features in the spatial patterns have significant contributions to the corresponding temporal patterns. This is done by identifying EEG features that have weights statistically significant different from zero. To determine the group-level spatial pattern, for each significant

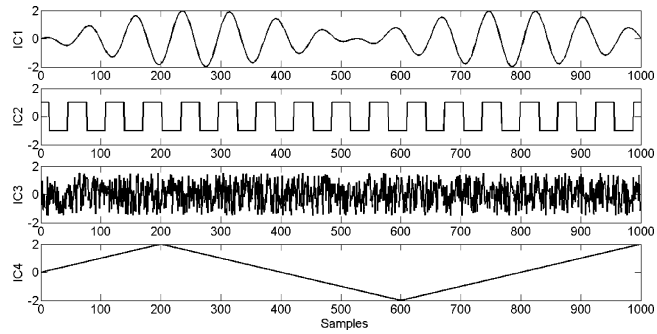

Fig. 2. Four source signals.

component, we applied a two-tailed t-test to each element of the spatial patterns of all subjects in each subject group.

III. Experiments and Results

A. Synthetic Data

In this simulation, we applied the proposed IC-PLS model to synthetic data and also implemented PLS and ICA for comparison. As an illustrative example, without loss of generality, four sources were generated and analyzed, similar to that used by Hakyin [26].

1) Data Generation: The following four source signals were considered:

$$\boldsymbol{s}_1 = 2\cos(0.08t)\sin(0.006t)$$

$$\boldsymbol{s}_2 = \text{sign}(\sin(0.3t) + 3\cos(0.1t))$$

$$\boldsymbol{s}_3 = \text{uniformly distributed noise in the range } [-1.5, 1.5]$$

$$\boldsymbol{s}_4 = \begin{cases} 0.01t & 1 \le t \le 200 \\ -0.01t + 4 & 201 \le t \le 600 \\ 0.01t + 8 & 601 \le t \le 1000 \end{cases} \tag{21}$$

where t denotes the time point index, valued from 1 to 1000, and $\boldsymbol{s}_i$s $(i = 1, 2, 3, 4)$ represented four simulated sources, as shown in Fig. 2.

Two mixed data sets $\boldsymbol{X}$ and $\boldsymbol{Y}$, shown in Fig. 3, were generated as follows:

$$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{s}_x \quad \text{and} \quad \boldsymbol{y} = \boldsymbol{B}\boldsymbol{s}_y \tag{22}$$

where $\boldsymbol{s}_x = [s_1\ s_2\ s_3]^T$ and $\boldsymbol{s}_y = [s_1\ s_2\ s_4]^T$ with

$$\boldsymbol{A} = \begin{bmatrix} 0.86 & 0.79 & 0.67 \\ -0.55 & 0.65 & 0.46 \\ 0.17 & 0.32 & -0.28 \\ -0.33 & 0.12 & 0.27 \\ 0.89 & -0.97 & -0.74 \end{bmatrix} \tag{23}$$

$$\boldsymbol{B} = \begin{bmatrix} 0.21 & 0.69 & 0.05 \\ 0.22 & 0.98 & 0.03 \\ 0.68 & 0.74 & 0.10 \\ 0.05 & 0.63 & 0.42 \\ 0.83 & 0.10 & 0.79 \end{bmatrix}. \tag{24}$$

Here, $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{s}_x$, and $\boldsymbol{s}_y$ are all column vectors, denoting one observation in their respective data space.

The sources $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$ exist in both data spaces $\boldsymbol{X}$ and $\boldsymbol{Y}$, representing common information. The source $\boldsymbol{s}_3$ only contributes to $\boldsymbol{X}$ and $\boldsymbol{s}_4$ only to $\boldsymbol{Y}$, representing their own unique information. In $\boldsymbol{Y}$, we intentionally assigned a relatively high weight to the source $\boldsymbol{s}_2$. Moreover, different white Gaussian noise with 5% power was added to each source in each data space.

B. Real EEG and EMG Data

We also applied the proposed IC-PLS group analysis framework for corticomuscular activity analysis to concurrent EEG and EMG signals collected from normal subjects and patient subjects with PD while they perform a visually guided tracking task. We aimed to extract common temporal patterns between EEG and EMG signals and explore the differences of the spatial patterns between the control and PD groups.

1) Data Collection and Preprocessing: The study was approved by the University of British Columbia Ethics Board, and all subjects gave written, informed consent prior to participating. Nine PD patients (mean age: 66 yrs) were recruited from the Pacific Parkinson's Research Centre at the University of British Columbia (Vancouver, Canada). They all displayed mild to moderate levels of PD severity (stages 1–2 on the Hoehn and Yahr scale) and were being treated with L-dopa medication (mean daily dose of 720 mg). All PD subjects were assessed after a minimum of 12-h withdrawal of L-dopa medication, and their motor symptoms were assessed using the Unified Parkinson's Disease Rating Scale, resulting in a mean score of 23. In addition, eight age-matched healthy subjects were recruited as controls. During the experiment, subjects seated 2-m away from a large computer screen. The visual target was displayed on the screen as a vertical yellow bar oscillating in height at 0.4 Hz. Subjects were asked to squeeze a pressure-responsive bulb with their right hand. The visual feedback representing the force output of the subject was displayed as a vertical green bar superimposed on the target bar as shown in Fig. 4. Applying greater pressure to the bulb increased the height of the green bar, and releasing pressure from the bulb decreased the height of the green bar. Subjects were instructed to make the height of the green bar match the height of target bar as closely as possible. Each squeezing period lasted for 15 s and was followed by a 15-s rest period. The squeezing task was performed twice. The force required was up to 10% of each subject's maximum voluntary contraction, which was measured at the beginning of each recording session.

The EEG data were collected using an EEG cap (QuickCap, Compumedics, Texas, USA) with 19 electrodes based on the International 10–20 system, referenced to linked mastoids. The EEG and EMG data were sampled at 1 kHz using SynAmps2 amplifiers (NeuroScan, Compumedics, Texas, USA). A surface electrode on the tip of the nose was used as ground. Ocular movement artifacts were measured using surface electrodes placed above and below the eyes (Xltek, Ontario, Canada). Data were later processed by a band-pass filter (1 to 70 Hz) offline and downsampled to 250 Hz. Artifacts associated with eye blinks and muscular activities were removed using the Automated Artifact Removal in the EEGLAB Matlab Toolbox

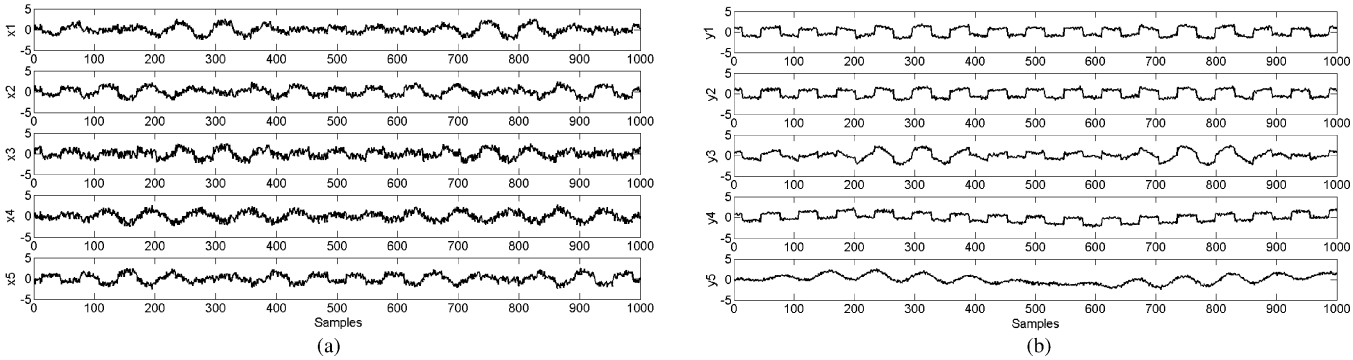

Fig. 3. (a) Mixed data X. (b) Mixed data Y.

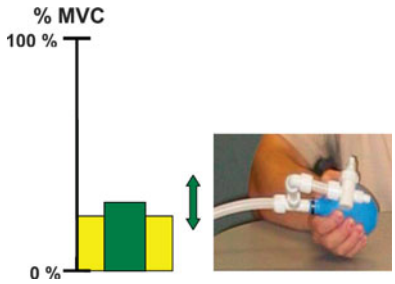

Fig. 4. Squeezing task: The subject was instructed to follow the target bar (yellow) as close as possible. The force exerted by the subject was shown by the green bar.

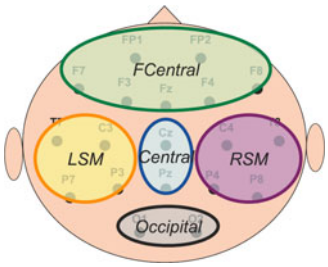

Fig. 5. Five brain regions: *FCentral* —FP1, FP2, F7, F3, Fz, F4 and F8, *LSM*—T7, C3, P7, and P3, *RSM*—T8, C4, P8 and P4, *Central* —Cz and Pz, and *Occipital*—O1 and O2.

(Gomez-Herrero, 2007). To simplify analysis, the EEG signals were divided into five regions as shown in Fig. 5. The five regions were: *Fronto-Central (FCentral)*—FP1, FP2, F7, F3, Fz, F4, and F8; *Left Sensorimotor (LSM)*—T7, C3, P7, and P3; *Right Sensorimotor (RSM)*—T8, C4, P8, and P4; *Central*—Cz and Pz; and *Occipital*—O1 and O2. While the proposed framework is capable of handling data at an individual electrode level, we found in our prior work [31] that averaging recordings within a region has several potential benefits: 1) it reduces the effects of possible spurious activity measured at individual electrodes; 2) it reduces intersubject variability due to slight variations in the placement of the electrode cap; and 3) regions can be chosen as they are biologically related to the motor task we are studying based on prior medical knowledge, making neuroscience interpretations easier. Thus, the raw time courses of the electrodes within each region were averaged as the overall activity of the region, and the averaged time course were then zero meaned and normalized to unit variance. For subsequent analysis, data collected during the squeezing periods were concatenated in time into a single matrix for each individual subject. Data from the rest periods were excluded from the analysis.

The EMG signals were recorded using self-adhesive, silver, silver-chloride pellet surface electrodes with 7-mm diameter. A bipolar montage was used with a fixed interelectrode distance of 30 mm. The surface EMG signals were simultaneously collected together with the EEG signals and were amplified and sampled at 1000 Hz. To be consistent with the EEG preprocessing, the EMG signals were downsampled offline to 250 Hz and only the squeezing periods are used for subsequent analysis.

We examined two types of EEG features: 1) the pairwise correlation between each individual region or 2) the pairwise coherence value between EEG channels in specific frequency bands. In both EEG feature settings, $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$ represent the time-varying feature matrices of EEG and EMG, respectively. The length of the original raw EEG and simultaneously recorded EMG data for each subject is 7485 associated with the sampling frequency 250 Hz. As mentioned before, after using the 300-length moving window and a 95% overlap, the length of time points (the matrix row) here is 480. For the EEG correlation case, since the EEG signals were divided into five regions as mentioned in Section III-B1, there are $C_2^5 = 10$ total correlation connections. Thus, $\boldsymbol{Z}_1$ is with size 480×10. For the EEG coherence case, there are five brain regions and four frequency bands, indicating there are totally $C_2^5 \times 4 = 40$ variables. Thus, $\boldsymbol{Z}_1$ is with size 480×40. For the EMG amplitude feature, since there are three surface EMG channels, $\boldsymbol{Z}_2$ was 480×3.

C. Results

1) Simulation Results: The extracted components using PLS, ICA, and the proposed IC-PLS model are shown in Figs. 6, 7, and 8 separately. The LVs extracted by PLS were automatically ordered in terms of their significance and successfully reflected the corresponding relationships of the underlying sources between $\boldsymbol{X}$ and $\boldsymbol{Y}$. However, compared with the original sources, the extracted LVs were distorted, suggesting that the fact that the PLS sources were uncorrelated was insufficient to accurately recover the underlying sources. In contrast, ICA recovered the original sources accurately in both data spaces,

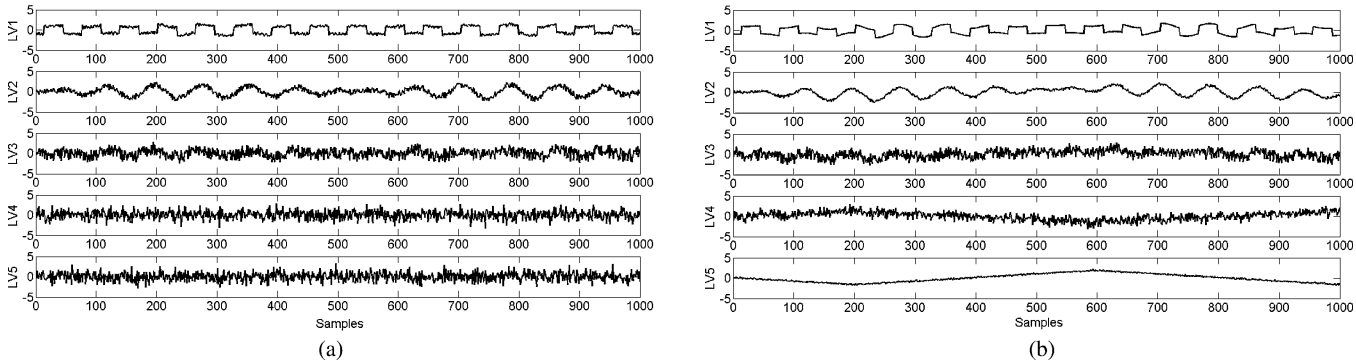

Fig. 6. (a) LVs estimated in *X* using PLS. (b) LVs estimated in *Y* using PLS.

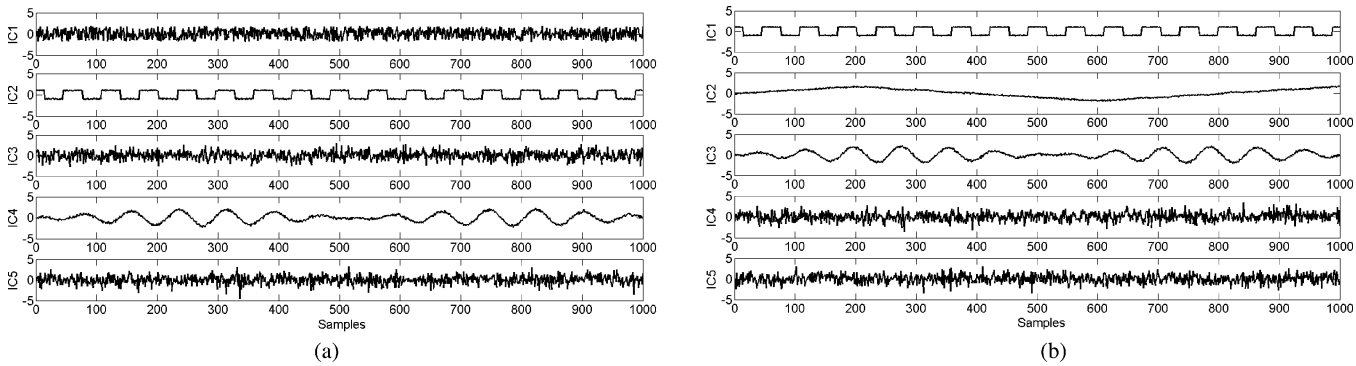

Fig. 7. (a) LVs estimated in *X* using ICA. (b) LVs estimated in *Y* using ICA.

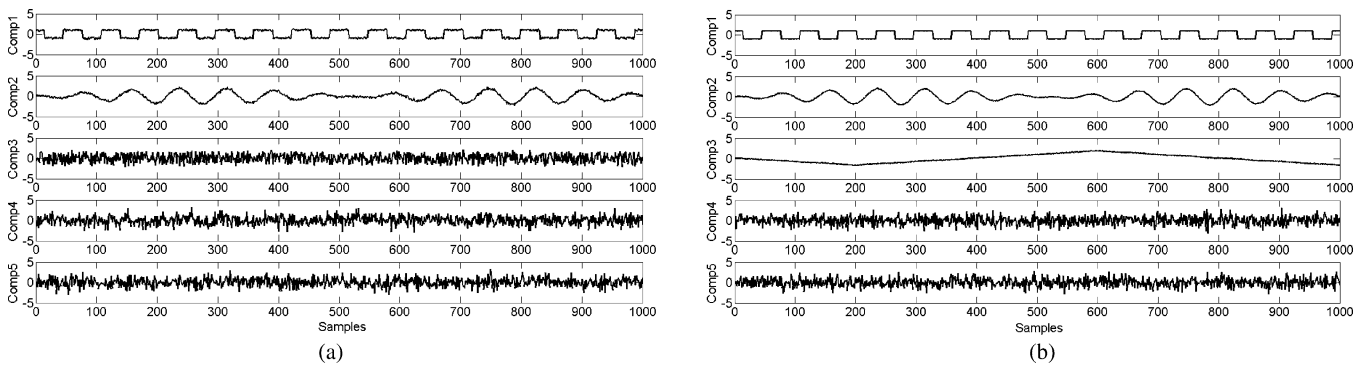

Fig. 8. (a) LVs estimated in *X* using IC-PLS. (b) LVs estimated in *Y* using IC-PLS.

but it did not relate the two data spaces and the LVs were unordered. When the proposed IC-PLS model was employed, the dominant sources that make significant contributions to both data spaces were accurately identified and ordered, with the focus entirely on sources that were common across the predictor and the response.

2) Results From Real Data: When the proposed group analysis framework was applied to EEG and EMG features generated using the procedure described in Section II-B2 from all eight normal and nine PD subjects simultaneously, the identification of common temporal patterns, but different spatial patterns could be identified. Using the permutation test, two components were deemed significant $(P \leq 0.05)$ (see Figs. 9 and 10). Note that in the figures only connections whose weights are statistically significantly different from zero are shown.

When using the correlation between brain regions as the EEG feature, many connections between were similar between PD and normals, especially when connections seen in either components were considered. The only difference was a connection between the *L* sensorimotor and frontal regions that was present in controls but missing in PD (see Fig. 9).

When the pairwise EEG coherence features were used to compare with the EMG features, there were also two components deemed significant via the permutation test (see Fig. 10). Again, when considering across components, there were similarities in the connections between PD and controls, but also significant differences. In the beta band, normal subjects had connections between the *R* sensorimotor cortex and occipital, central, and frontal regions that were completely missing in PD subjects. In the gamma band, PD subjects had increased connectivity

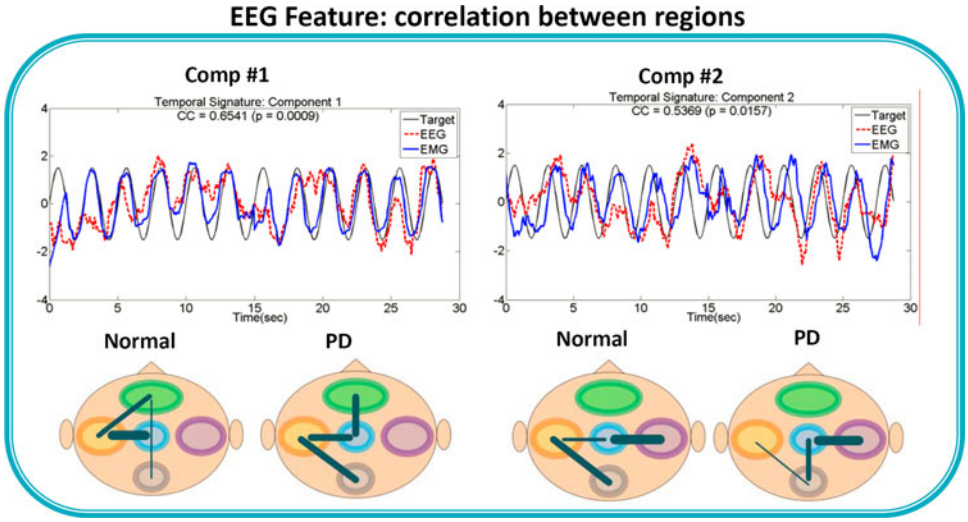

Fig. 9. Two components of the group analysis framework when using the EEG correlations and the EMG amplitude data sets. Top panel: Temporal patterns of the EEG (red, dashed) and the EMG (blue, solid). The oscillation of the target bar is also shown (black, solid). Bottom panel: EEG spatial patterns of normal subjects (left) and PD subjects (right). The connections reflect the respective spatial patterns in the two groups and the width is made proportional to the weighting of the respective connections in the group EEG spatial pattern. CC means correlation coefficient.

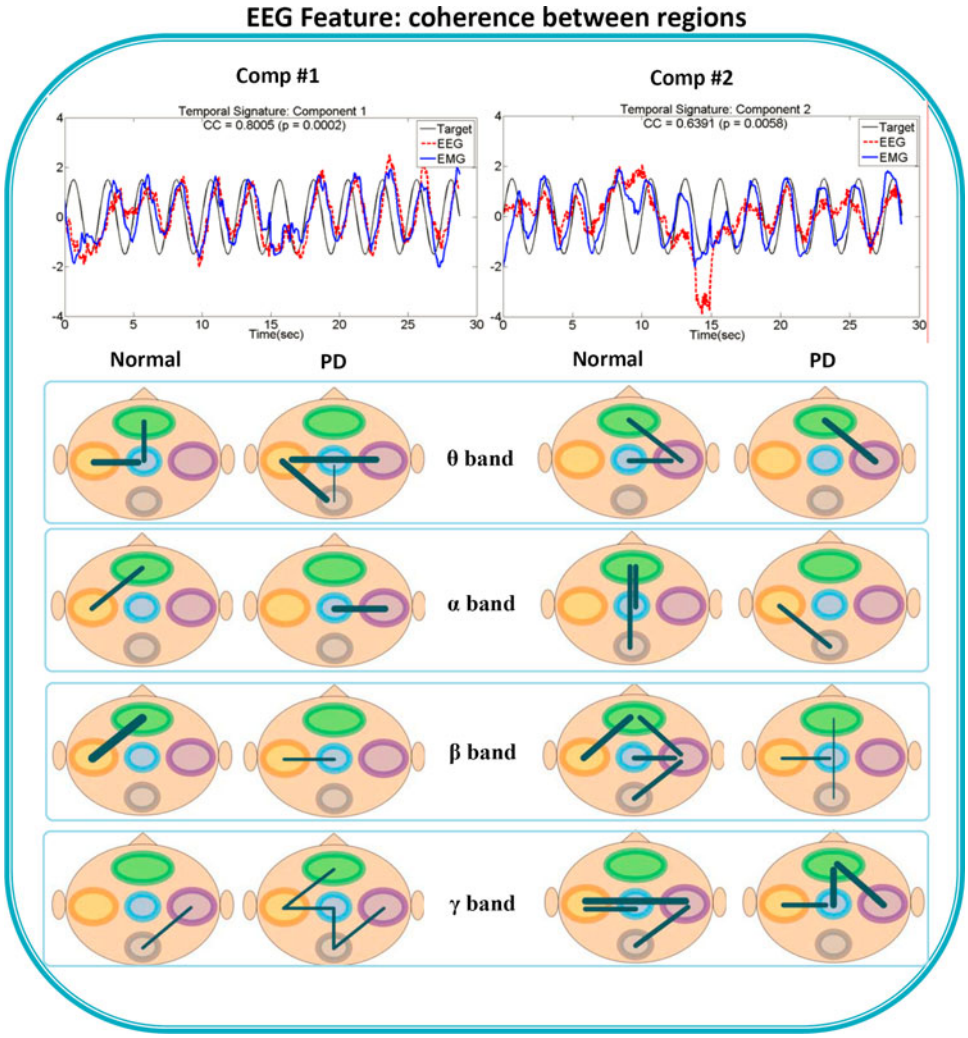

Fig. 10. Two components of the group analysis framework when using the EEG coherence and the EMG amplitude data sets.

between the frontal regions and central and sensorimotor cortices that were missing in controls. Finally, PD subjects had a significant connection between the *L* sensorimotor and occipital regions in the theta and alpha bands that was not present in controls.

IV. Discussion

In this paper, we have proposed a new method to assess EEG/EMG coupling based on combining the favorable properties of ICA and PLS. The simulations support utilization of proposed method, and results based on real data from Parkinson's and normal control subjects performing a dynamic motor task are promising. In the past, most EEG/EMG coupling studies have compared EEG activity at a specific locus (e.g., sensorimotor cortex contralateral to the hand performing the task) with the EMG during sustained contraction. However, we found that in normal subjects, correlation between the contralateral sensorimotor cortex and other regions was closely associated with ongoing EMG features during dynamic motor performance (see Fig. 9). It is likely that the dynamic nature of the task might require the recruitment of additional regions such as frontal regions for motor selection [28], contralateral (i.e., ipsilateral to hand movement) sensorimotor cortex for fine modulatory control [29], and occipital regions for posterror adaptations [30].

The results we observed in the PD and control subjects are consistent with prior studies describing the changes in EEG observed in PD. When using the correlation between brain regions as EEG features, we found widespread connectivity between central regions and the ipsilateral (right) sensorimotor regions and ongoing EMG activity. A body of literature suggests that widespread oscillatory disruption in PD is detectable in the EEG during visually guided tasks (e.g., [31]). This likely represents compensatory expansion of cortical regions to overcome deficiencies in the basal ganglia to perform motor tasks, as has been observed in fMRI studies [32]. The lack of FCentral alpha activity seen in PD subjects compared to controls may reflect impaired spatial attention [33], as impairment of maintenance of attention has been well described in PD subjects (e.g., [34]). The increased frontal connectivity we observed in the gamma region in PD may reflect the enhanced effort PD subjects required for motor task switching [35], a problem common in this population [36]. The enhanced occipital connectivity seen in PD in the theta and beta bands likely relates to the fact that PD subjects heavily rely on visual cues for the initiation [37] and ongoing control of movement [38].

It might seem paradoxical that normal subjects had enhanced beta-band connectivity in the sensorimotor cortex ipsilateral to the squeezing hand, especially since Parkinson's subjects are thought to have enhanced beta-band EEG activity [39]. However, it is important to remember that the current method specifically looks at the common sources in the EEG and EMG. Since a key aspect in PD is the inability to modulate the beta-band oscillations [40], it is likely that the fraction of beta-band activity that varies with phasic EMG activity is reduced in PD.

We were able to find common temporal patterns of corticomuscular coupling in both PD and controls despite having different spatial origins of this common signal. We note that our results are consistent with a recent study that suggests the magnitude of traditional MSC is not significantly different early in PD [41]. However, we have shown that the distributed EEG connectivity patterns required to generate these common temporal patterns differ between PD and controls. One could envision situations where the spatial extent is similar, but the temporal profiles are significantly different (e.g., focal stroke affecting the sensorimotor cortex contralateral to the hand performing the task) which may not be well served by the proposed approach. However, even in this scenario, compensatory mechanisms would still presumably modify the connectivity patterns giving rise to corticomuscular coherence, and this would still be captured by the current approach.

A limitation of our approach is that it treats corticomuscular interactions as bidirectional; however, in addition to the cortex driving the muscle, there may be proprioceptive information from the muscle to the cortex. This reflects the inherent limitation of the PLS step; however, recent work on directed PLS (e.g., [42]) may provide future directions to extend the current approach. Another possible refining is that the cross validation and bootstrap method may be used to test the robustness and infer the population level information. We note that while the proposed framework is proposed for corticomuscular coupling analysis, it can be generally applied to other types of concurrent signals, including, but not limited to, fMRI, PPG, ECG, and kinematic data. Therefore, the proposed framework is a promising technique for performing multisubject and bimodal data analysis.

Appendix

In this appendix, we show how to mathematically derive the optimization solution of the proposed IC-PLS model.

By employing the method of Lagrange multipliers, we can rewrite the initial cost function as follows:

$$\begin{aligned}\mathrm{F}(\boldsymbol{w}_1, \boldsymbol{w}_2, \lambda_1, \lambda_2) = {} & \alpha \left(\mathrm{E}((\boldsymbol{w}_1{}^T \boldsymbol{x})(\boldsymbol{w}_2{}^T \boldsymbol{y}))\right)^2 \\ & + \beta \left(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T \boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1))\right)^2 \\ & + \theta \left(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T \boldsymbol{y})) - \mathrm{E}(\mathrm{G}(u_2))\right)^2 \\ & + \lambda_1 (\boldsymbol{w}_1{}^T \boldsymbol{w}_1 - 1) + \lambda_2 (\boldsymbol{w}_2{}^T \boldsymbol{w}_2 - 1) \end{aligned} \quad \text{(A-1)}$$

where λ_1 and λ_2 are Lagrange multipliers.

Here, we only present the detailed derivations regarding $\boldsymbol{w}_1$. As to those of $\boldsymbol{w}_2$, it is straightforward from the results obtained on $\boldsymbol{w}_1$. Taking the derivatives of $\mathrm{F}(\boldsymbol{w}_1, \boldsymbol{w}_2, \lambda_1, \lambda_2)$ with respect to $\boldsymbol{w}_1$ and λ_1 and setting them to be zero, we have

$$\begin{aligned}\nabla \mathrm{F}_{\boldsymbol{w}_1} = \frac{\partial \mathrm{F}}{\partial \boldsymbol{w}_1} = {} & 2\alpha \mathrm{E}((\boldsymbol{w}_1{}^T \boldsymbol{x})(\boldsymbol{w}_2{}^T \boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T \boldsymbol{y})) \\ & + 2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T \boldsymbol{x})) - \mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T \boldsymbol{x})) \\ & + 2\lambda_1 \boldsymbol{w}_1 = 0, \end{aligned} \quad \text{(A-2)}$$

$$\nabla \mathrm{F}_{\lambda_1} = \boldsymbol{w}_1{}^T \boldsymbol{w}_1 - 1 = 0 \quad \text{(A-3)}$$

where $\mathrm{g}(\cdot)$ represents the corresponding first-order derivative of $\mathrm{G}(\cdot)$.

Left multiplying both sides of (A-2) by ${\boldsymbol{w}_1}^T$, we have

$$\begin{aligned}&2\alpha {\boldsymbol{w}_1}^T \mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&\quad+2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\boldsymbol{w}_1{}^T\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&\quad+2\lambda_1\boldsymbol{w}_1{}^T\boldsymbol{w}_1=0.\end{aligned}\tag{A-4}$$

According to (A-3), λ_1 can be calculated as

$$\begin{aligned}\lambda_1=&-\alpha\boldsymbol{w}_1{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&-\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\boldsymbol{w}_1{}^T\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x})).\end{aligned}\tag{A-5}$$

Based on Kuhn–Tucker conditions [27], the optima of the multiobjective function shown in (5) will be at the points shown in (A-2) under the constraint (A-3). In this study, to improve the convergence speed, Newton's method is employed to solve this problem.

Suppose $\Phi_1=\nabla\mathrm{F}_{\boldsymbol{w}_1}$ and then its Jacobian matrix can be derived as

$$\begin{aligned}\mathrm{J}\Phi_1=&\,2\alpha\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}^T(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\boldsymbol{x}\boldsymbol{x}^T\mathrm{g}'(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&+2\beta\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\mathrm{E}(\boldsymbol{x}^T\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))+2\lambda_1\mathrm{I}\end{aligned}\tag{A-6}$$

where $\mathrm{g}'(.)$ means the second-order derivative of G(.).

Substituting (A-5) into (A-6), we have the following expression for $\mathrm{J}\Phi_1$:

$$\begin{aligned}\mathrm{J}\Phi_1=&\,2\alpha\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}^T(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\boldsymbol{x}\boldsymbol{x}^T\mathrm{g}'(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&+2\beta\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\mathrm{E}(\boldsymbol{x}^T\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&-(2\alpha\boldsymbol{w}_1{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\boldsymbol{w}_1{}^T\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x})))\mathrm{I}.\end{aligned}\tag{A-7}$$

Since the data have been initialized, to simplify the inverse of the Jacobian matrix, the second term can be approximated as in the original FastICA algorithm as [13]

$$\mathrm{E}(\boldsymbol{x}\boldsymbol{x}^T\mathrm{g}'(\boldsymbol{w}_1{}^T\boldsymbol{x}))\approx\mathrm{E}(\boldsymbol{x}\boldsymbol{x}^T)\mathrm{E}(\mathrm{g}'(\boldsymbol{w}_1{}^T\boldsymbol{x}))=\mathrm{E}(\mathrm{g}'(\boldsymbol{w}_1{}^T\boldsymbol{x}))\mathrm{I}.\tag{A-8}$$

Therefore, the Jacobian matrix can be approximately expressed in a simple form

$$\begin{aligned}\mathrm{J}\Phi_1=&\,2\alpha\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}^T(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+2\beta\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\mathrm{E}(\boldsymbol{x}^T\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-c_1\mathrm{I}\end{aligned}\tag{A-9}$$

where c_1 is a constant defined as

$$\begin{aligned}c_1=&\,2\alpha\boldsymbol{w}_1{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\boldsymbol{w}_1{}^T\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&-2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\mathrm{g}'(\boldsymbol{w}_1{}^T\boldsymbol{x})).\end{aligned}\tag{A-10}$$

Substituting (A-5) into (A-2), we can express the derivatives of $\mathrm{F}(\boldsymbol{w}_1,\boldsymbol{w}_2,\lambda_1,\lambda_2)$ with respect to $\boldsymbol{w}_1$ as

$$\begin{aligned}\nabla\mathrm{F}_{\boldsymbol{w}_1}=&\,2\alpha\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+2\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&-2(\alpha\boldsymbol{w}_1{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{x}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&+\beta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_1{}^T\boldsymbol{x}))-\mathrm{E}(\mathrm{G}(u_1)))\boldsymbol{w}_1{}^T\mathrm{E}(\boldsymbol{x}\mathrm{g}(\boldsymbol{w}_1{}^T\boldsymbol{x})))\boldsymbol{w}_1.\end{aligned}\tag{A-11}$$

Through the similar procedure, $\mathrm{J}\Phi_2$ and $\nabla\mathrm{F}_{\boldsymbol{w}_2}$ can be easily derived as

$$\begin{aligned}\mathrm{J}\Phi_2=&\,2\alpha\mathrm{E}(\boldsymbol{y}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\mathrm{E}(\boldsymbol{y}^T(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&+2\theta\mathrm{E}(\boldsymbol{y}\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{y}^T\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y}))-c_2\mathrm{I}\end{aligned}\tag{A-12}$$

where c_2 is a constant defined as

$$\begin{aligned}c_2=&\,2\alpha\boldsymbol{w}_2{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{y}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&+2\theta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y}))-\mathrm{E}(\mathrm{G}(u_2)))\boldsymbol{w}_2{}^T\mathrm{E}(\boldsymbol{y}\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&-2\theta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y}))-\mathrm{E}(\mathrm{G}(u_2)))\mathrm{E}(\mathrm{g}'(\boldsymbol{w}_2{}^T\boldsymbol{y}))\end{aligned}\tag{A-13}$$

and

$$\begin{aligned}\nabla\mathrm{F}_{\boldsymbol{w}_2}=&\,2\alpha\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{y}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&+2\theta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y}))-\mathrm{E}(\mathrm{G}(u_2)))\mathrm{E}(\boldsymbol{y}\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y}))\\&-2(\alpha\boldsymbol{w}_2{}^T\mathrm{E}((\boldsymbol{w}_1{}^T\boldsymbol{x})(\boldsymbol{w}_2{}^T\boldsymbol{y}))\mathrm{E}(\boldsymbol{y}(\boldsymbol{w}_1{}^T\boldsymbol{x}))\\&+\theta(\mathrm{E}(\mathrm{G}(\boldsymbol{w}_2{}^T\boldsymbol{y}))-\mathrm{E}(\mathrm{G}(u_2)))\boldsymbol{w}_2{}^T\\&\times\mathrm{E}(\boldsymbol{y}\mathrm{g}(\boldsymbol{w}_2{}^T\boldsymbol{y})))\boldsymbol{w}_2.\end{aligned}\tag{A-14}$$

The Newton iteration direction vectors $\boldsymbol{d}_i(i=1,2)$ are derived by solving the following equations:

$$\mathrm{J}\Phi(\boldsymbol{w}_i)\cdot\boldsymbol{d}_i=-\nabla\mathrm{F}_{\boldsymbol{w}_i},\quad i=1,2.\tag{A-15}$$

Therefore, finally, $\boldsymbol{w}_i(i=1,2)$ can be derived as follows:

$$\begin{aligned}&\boldsymbol{w}_i\leftarrow\boldsymbol{w}_i+\boldsymbol{d}_i,\\&\text{i.e.,}\quad\boldsymbol{w}_i\leftarrow\boldsymbol{w}_i-(\mathrm{J}\Phi(\boldsymbol{w}_i))^{-1}\nabla\mathrm{F}_{\boldsymbol{w}_i},\quad i=1,2.\end{aligned}\tag{A-16}$$

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